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CSA 5111

CIA TEST-1

1. plaintext : MEETAT NOON

Shift : 7

encryption : $C = (P+7) \bmod 26$

M	E	E	T	A	T		N	O	O	N
13	5	5	20	1	20		14	15	15	14

1. $C = (13+7) \bmod 26 = 20$
2. $C = (5+7) \bmod 26 = 12$
3. $C = (5+7) \bmod 26 = 12$
4. $C = (20+7) \bmod 26 = 1$
5. $C = (1+7) \bmod 26 = 8$
6. $C = (20+7) \bmod 26 = 1$
7. $C = (14+7) \bmod 26 = 21$
8. $C = (15+7) \bmod 26 = 22$
9. $C = (15+7) \bmod 26 = 22$
10. $C = (14+7) \bmod 26 = 21$

C : 20 12 12 1 8 1 21 22 22 21

Cipher text : T L L A H A U V V U

∴ cipher text : T LL A H A U V V U

Decryption :

$$P = (C - 7) \bmod 26$$

1. $P = (20 - 7) \bmod 26$

$= 13$

2. $P = (18 - 7) \bmod 26 = 5$

P : 13 5 5 20 1 20 14 15 15 14

Plain text : M E E T A T N O O N

(b) Vigenère cipher

plaintext : MEET AT NOON

$$c = (p+K) \bmod 26$$

key = SUN

M E E T A T N O O N
S U N S U N S U N S

M E E T A T N O O N
13 5 5 20 1 20 14 15 15 14

S	U	N	S	U	N	S	U	N	S
19	21	14	19	21	14	19	21	14	19

encryption text :

1. $c = (p+K) \bmod 26 \Rightarrow (13+19) \bmod 26$
$= 6$

2. $c = (p+K) \bmod 26 \Rightarrow (5+21) \bmod 26$
$= 25$

3. $c = (p+K) \bmod 26 \Rightarrow (5+14) \bmod 26$
$= 19$

4. $c = (p+k) \bmod 26 \Rightarrow (20+19) \bmod 26$

$= 13$

5. $c = (p+k) \bmod 26 \Rightarrow (1+21) \bmod 26$

$= 22$

6. $c = (p+k) \bmod 26 \Rightarrow (20+14) \bmod 26$

$= 14$

7. $c = (p+k) \bmod 26 \Rightarrow (14+19) \bmod 26$

$= 7$

8. $c = (p+k) \bmod 26 \Rightarrow (15+21) \bmod 26$

$= 36 \bmod 26 = 10$

9. $c = (p+k) \bmod 26 \Rightarrow (15+4) \bmod 26$

$\Rightarrow 29 \bmod 26 = 3$

10. $c = (p+k) \bmod 26 \Rightarrow (14+19) \bmod 26$

$= 33 \bmod 26$

$= 7$

plain text : M E E T A T N O O N

c : 6 25 19 13 22 14 7 10 3 7

cipher text : F Y S M V N G J C G

Decryption (vignere cipher)

$$P = (C - K) \bmod 26$$

1. $P = (6 - 19) \bmod 26 = 13 \Rightarrow M$
2. $P = (25 - 21) \bmod 26 = 5 \Rightarrow E$
3. $P = (18 - 14) \bmod 26 = 5 \Rightarrow E$
4. $P = (13 - 19) \bmod 26 = 20 \Rightarrow T$
5. $P = (22 - 21) \bmod 26 = 1 \Rightarrow A$
6. $P = (24 - 14) \bmod 26 = 20 \Rightarrow T$
7. $P = (14 - 19) \bmod 26 = 14 \Rightarrow N$
8. $P = (10 - 21) \bmod 26 = 15 \Rightarrow O$
9. $P = (3 - 14) \bmod 26 = 15 \Rightarrow O$
10. $P = (4 - 19) \bmod 26 = 14 \Rightarrow N$

(c)

ceaser cipher	Vignère cipher
(i) Keyspace size is small	(ii) Keyspace size is large compared to ceaser
(iii) For same letter there will be same cipher	(iii) For same letter there will be be different

(iii) Broken by brutal force	(iii) strong enough than caesar
(iv) vulnerable to cryptanalysis	(iv) less vulnerable than caesar

Therefore, vigenère cipher is better than caeser's cipher

But for modern world, AES-256-GCM is better than vigenère cipher and caeser cipher.

Because,

(i) Both are lack of Integrity

(ii) less Protection for Confidentiality of data

(iii) Both are vulnerable to cryptanalyze

2. HILL CIPHER

Plaintext = MATH

Key Matrix :

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

M	A	T	H
12	0	19	7

$$P_1 : MA = \begin{bmatrix} 12 \\ 0 \end{bmatrix}$$

$$P_2 : TH = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$$

Encryption : $C = KP \bmod 26$

$$C_1 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} = \begin{bmatrix} (36+0) \\ (60+0) \end{bmatrix} = \begin{bmatrix} 36 \\ 60 \end{bmatrix}$$

$\bmod 26$

$$= \begin{bmatrix} 10 \\ 8 \end{bmatrix}$$

$10 = K$, $8 = I$

First ciphertext = KI

For P2,

$$C_2 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} = \begin{bmatrix} 71 \\ 144 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$19 = T$, $14 = O$

Second ciphertext = TO

C. Check whether the key matrix is Invertible

Determinant :

$$|K| = (a \times d) - (b \times c)$$

$$= (3 \times 7) - (2 \times 5)$$

$$= 21 - 10 = 11$$

$$\gcd(11, 26) = 1$$

Therefore, the key matrix is invertible Modulo 26

D. Modular einverse of determinant

$11 \times 19 = 209$ [$209 = 26 \times 8 + 1$]

$209 \bmod 26 = 1$

$11^{-1} = 19 \bmod 26$

E. Find inverse key matrix

$$adj(k) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \Rightarrow \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$k^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26$$

$133 \bmod 26 = 3$
$-38 \bmod 26 = 14$
$-95 \bmod 26 = 9$
$57 \bmod 26 = 5$

$$\therefore k^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$kk^{-1} \equiv I \pmod{26}$$

$$= \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} = \begin{bmatrix} 27 & 52 \\ 78 & 105 \end{bmatrix} \bmod 26 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Decryption:

$$P = k^{-1} C \bmod 26$$

$$k^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$C_1 = \begin{bmatrix} 10 \\ 8 \end{bmatrix}, \quad C_2 = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} = \begin{bmatrix} (3\times10) + (14\times8) \\ (9\times10) + (5\times8) \end{bmatrix}$$

$$= \begin{bmatrix} 142 \\ 130 \end{bmatrix} \bmod 26 = \begin{bmatrix} 12 \\ 0 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 12 \\ 0 \end{bmatrix} \qquad KI \rightarrow MA$$

$$P_2 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{bmatrix} (3\times19) + (14\times14) \\ (9\times19) + (5\times14) \end{bmatrix}$$

$$= \begin{bmatrix} 253 \\ 241 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$$

$$P_2 = \begin{bmatrix} 19 \\ 7 \end{bmatrix} \qquad TO \rightarrow TH$$

Weakness of hill cipher :

1. Hill cipher uses linear mathematical operation for encryption.
2. It can't detect unauthorized modification of encrypted data.
3. doesn't verify the identity of sender

Q3 AES-256-GCM Vs RSA and Hybrid Encryption

Comparision of AES and RSA

AES is a Symmetric encryption algorithm whereas RSA is an assymmetric algorithm

Feature	AES-256-GCM	RSA
Key usage	Same Secret key for encryption and Decryption	Public and Private keys
Speed	very fast	slower
Data size	Suitable for large files	Suitable for Small data
Key size	256-bit key	2048 bit modules

1. Key usage
   - AES uses one shared Secret keys
   - RSA uses Public and Private keys

2. Performance
   - AES is fast and suitable for large files
   - RSA is slower and suitable for small data

3. Scalability :

- 200 clinics requires 19,900 shared keys

$$\frac{n(n-1)}{2}$$

$$\therefore \frac{200 \times 199}{2} = 19,900$$

Receiving Server :

- Decrypt the MRI file using DEK and nonce
- verify the authentication tag before accepting the file.

Secure key storage :

- store keys using HSM / KMS
- use access control and MFA

Key Rotation :

- Generate a separate DEK for each file.
- periodically rotate encryption keys

4.

Prime number, $p = 11$

$q = 19$

$e = 7$

$M = 5$

(1) Find modulus

$n = p \times q$

$= 11 \times 19$

$= 209$

$\boxed{n = 209}$

11×19

99

114

209

(2) Find euler's totient function

$\phi(n) = (p-1)(q-1)$

$= (11-1)(19-1)$

$= 10 \times 18 = 180$

$\boxed{\phi(n) = 180}$

(3) GCD

$GCD \equiv (e, \phi(n)) \bmod n$

$\equiv (7, 180) \bmod 209$

$\equiv 1$

∴ It is invertible

Public key : $(e, n) = (7, 209)$

Private key : $e \times d \equiv \mod n \Rightarrow 7d \equiv \mod 180$

$$180 = 7 \times 25 + 5$$
$$7 = 5 \times 1 + 2$$
$$5 = 2 \times 2 + 0$$

Back substitution,

$$1 = 5 - 2 \times 2$$
$$\boxed{2 = 7 - 5}$$
$$1 = 5 - (7 - 5) \times 2$$
$$1 = 3(5) - 2(7)$$
$$\boxed{5 = 180 - 7(25)}$$

$-77 \times 7 \equiv 1 \pmod{180}$

$d = -77 \mod 180$

$\boxed{d = 103}$

$25 \times 25 = 625$

RSA encryption :

$$C = M^e \mod n$$
$$= 5^7 \mod 209$$
$$5^7 = 5^4 * 5^2 * 5 \mod 209$$
$$= 625 * 25 \times 5 \mod 209$$

$625 \mod 209 = 209$

$$= 125 \times 207 \mod 209$$
$$= 2 \times 84$$

$\boxed{C = 168}$

$209 \overline{)625}\ (3$

$209 \overline{)416}\ (2$

$209 \overline{)207}$

RSA Decryption :

$M = c^d \bmod n$

$M = 168^{103} \bmod 209$

$103 = 64 + 32 + 4 + 2 + 1$

$= 103^{64} + 103^{32} + 103^4 + 103^2 + 103^1$

$\boxed{M = 5}$

Private key, $(\vec{d}, \vec{n}) = (103, 209)$

Types of cryptography techniques

1. Caeser cipher
2. Playfair cipher
3. Hill cipher
4. RSA - 256

1. caeser cipher is a substitution technique used to convert a plaintext to encryption and ciphertext to original

Plaintext : MEETATNOON

Shift 7 :

encryption : $c = (p+7) \bmod 26$

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13	5	5	20	1	20	14	15	15	14

1. $c = (13+7) \bmod 26 = 20$
2. $c = (5+7) \bmod 26 = 12$
3. $c = (5+7) \bmod 26 = 12$
4. $c = (20+7) \bmod 26 = 1$

5. $c = (1+7) \bmod 26 = 8$

6. $c = (20+7) \bmod 26 = 1$

7. $c = (14+7) \bmod 26 = 21$

8. $c = (15+7) \bmod 26 = 22$

9. $c = (15+7) \bmod 26 = 22$

10. $c = (14+7) \bmod 26 = 21$

c = 20 12 12 1 8 1 21 22 22 21

cipher text : T L L A H A U V V U

∴ cipher text : T L L A H A U V V U

Decryption :

$$p = (p-7) \bmod 26$$

p = 13 5 5 20 1 20 14 15 15 1

original text : M E E T A T N O O N

HILL CIPHER

plaintext = MATH

$$key = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

M A T H

12 0 19 7

Encryption : $C = kp \bmod 26$

$$C_1 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \end{bmatrix} \Rightarrow KI$$

$$C_2 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} = \begin{bmatrix} 71 \\ 144 \end{bmatrix} \Rightarrow TO$$

Decrypt $= k^{-1} \bmod 26$

$$K_1 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \end{bmatrix} \Rightarrow MA$$

$$K2 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{bmatrix} 19 \\ 7 \end{bmatrix} \Rightarrow TH$$

Encryption cypes text : KITO

Decrypt text : MATH

AES
~~RSA~~ - 256 - GCM is the technique which suits more than caeser cipher, playfair cipher, hill cipher

Because, for Protecting Sensitive financial data, AES - 256 - GCM is best